Evaluating the Robustness of Transformer-based Models Against Data Poisoning Attacks in Social Media Sentiment Analysis

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Abstract—Transformer-based Pre-trained Language Models (PLMs) has become standard tools for sentiment analysis on social platforms, yet their vulnerability to data poisoning during training remains a critical concern. In this paper, we conduct a baseline resilience study of BERT, RoBERTa, and BERTweet against three increasingly complex adversarial attacks: label flipping, trigger injection, and clean-label poisoning. Using the Sentiment140 corpus, we introduced poisoned data at rates of 5%, 10%, 20% and 30%, tracking performance across multiple metrics including Accuracy, Macro-F1, Attack Success Rate (ASR), and Clean Accuracy (CACC). The experimental data—averaged over three random seeds—revealed that while basic label flipping caused steady performance degradation, our conservative configurations of trigger injection and clean-label poisoning often failed to disrupt the models reliably. Ultimately, the domain-aligned BERTweet model demonstrated the highest baseline resilience to these specific manipulations, highlighting the need for future evaluations using more advanced threat models.

Keywords—data poisoning, transformer models, sentiment analysis, BERT, RoBERTa, BERTweet, adversarial robustness, backdoor attack, label flipping, clean-label attack, social media NLP.

I. INTRODUCTION

Automated sentiment analysis of social media content has become a core component of opinion mining, brand monitoring, and policy research. The scale of user-generated content on platforms such as Twitter makes manual annotation and labelling of them not practical, puts NLP models that can handle informal, noisy text in very high demand. Pre-trained Language Models (PLMs) grounded in the Transformer architecture BERT [1] and its successors [2], [3] have established strong results on sentiment classification and related tasks [22], [23], and are routinely deployed in production pipelines that continuously trained on fresh data.

This continuous retraining loop creates an exploitable vulnerability. Data poisoning [4], [25] is a training-time threat in which an adversary injects a small fraction of manipulated sets of samples into the corpus, with the goal of degrading accuracy or introducing hidden misclassification behaviour [5], [18]. Because scraped social media data cannot be reliably marked before use, attacks of that nature are difficult to detect and practical.

While existing literature documents the impact of isolated attack vectors on specific models [5], [8], [12], it lacks a unified, comparative assessment. Specifically, current benchmarks fail to cover: (a) diverse transformer variants of architectures varying in pre-training volume and domain specialization; (b) multiple concurrent poisoning strategies; and (c) evaluations conducted exclusively on authentic social media text [3], [26]. To bridge this gap, three research questions are framed around this work. RQ1 asks how each data poisoning techniques—label flipping, trigger injection, and clean-label—in varying degrees of poisoning rates affects BERT, RoBERTa, and BERTweet. RQ2 examines whether larger pre-training scale (RoBERTa) or domain-specific pre-training (BERTweet) provides significant robustness advantages over the BERT baseline. RQ3 determines which strategy poses the greatest practical threat to transformer-based classifiers on real-world social media text.

The study contributes (i) a systematic comparative evaluation of standard, optimized, and domain-specific Transformer architectures under data poisoning on social media text; and (ii) a rigorous robustness analysis across three escalating attack categories, with results reported as mean ± standard deviation across three seeds.

To bridge this gap, this work is explicitly framed as a baseline resilience study rather than a comprehensive threat assessment. We define a realistic threat model assuming an attacker who can inject a limited amount of poisoned data into automated social media scraping pipelines, but lacks access to the model's internal weights or defense mechanisms. While practical in real-world deployments, this restricts the adversary to black-box data poisoning.

II. RELATED WORKS

A. Sentiment Analysis on Social Media

Sentiment analysis identifies the writer's attitude toward a subject [1], [22]. On social media, informal language—slang, hashtags, abbreviations, character-level variations—substantially degrades lexicon-based and bag-of-words approaches [3], [23], [26]. Transformer PLMs have raised the

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performance ceiling while simultaneously exposing new risks around training-data integrity [5], [18].

B. Transformer Architectures

Three architectures represent distinct pre-training paradigms [20]. BERT, a bidirectional Transformer variant comprising 12 encoder layers, a 768-dimensional hidden state, and ≈110M parameters, originally trained via masked language modelling and next-sentence prediction objectives [1]. RoBERTa [2] refines BERT with a larger corpus (>160 GB), dynamic masking, larger batches, and no NSP objective, yielding consistent gains across NLP benchmarks. To address the linguistic traits of social media text, BERTweet adapted these identical RoBERTa optimization conditions to a specialized dataset of 850 million English tweets [3]. The resulting domain-specific vocabulary enables more precise modelling of platform-specific language, consistently outperforming general-domain encoders on uncurated tweet data.

C. Data Poisoning Attacks

The vulnerability of language models to training-time manipulation is well-documented, with adversaries injecting corrupted samples to either degrade global accuracy or implant hidden, conditional behaviours [4], [25]. Literature categorizes these threat vectors into three strategies of escalating complexity. Label flipping flips the ground-truth target values for a subset of the training data but leaves the input text unchanged [4], [14]. Trigger injection low- pairs trigger tokens with modified labels in some training sets [5], [27]. This forces the model to learn a latent association between the trigger and label association shows the expected baseline performance on clean datasets but only triggers the malicious response at inference when the rare token is present. As a stealthier alternative to automated label audits and data cleaning pipelines [6], [19], clean-label poisoning was proposed. Instead of manipulating the labels, this technique adds imperceptible textual perturbations (e.g., sentiment-weakening synonym substitutions, or typographic, character-level noise), thereby misaligning the input features and the clean label.

D. Defense Mechanisms and Research Gap

There are usually a few different types of defensive frameworks that are used to combat these vectors. Certified defenses [4] provide mathematical robustness guarantees against bounded poisoning rates. Alternatively, proactive measures, such as outlier detection and influence-function analysis, attempt to flag and isolate poisoned instances before training, but their effectiveness diminishes significantly against covert clean-label attacks [6]. Some defenses attempt to contain backdoors by looking for anomalies in intermediate representations, e.g. activation clustering [31] and spectral signatures [32]. However, defenses are known to be brittle against adaptive or very subtle triggers [29]. Importantly, as summarized in Table I, the existing literature does not have a unified benchmark that evaluates all the three attack vectors at once on BERT, RoBERTa and BERTweet in a real-world social media corpus.

TABLE I. COMPARISON OF RELATED STUDIES ON DATA POISONING IN NLP

Study	Models	Attack Types	Dataset	Multi-model Comparison
Kurita et al. [5]	BERT	Backdoor (trigger)	SST-2, IMDB	No
Wallace et al. [6]	BERT, ELMo	Universal adversarial triggers	Multiple NLP	Partial
Steinhardt et al. [4]	SVM, MLP	Label flipping	Enron, MNIST	No
Zhang et al. [9]	CNN, RNN, BERT	Survey (multiple)	Various	Partial
This study	**BERT, RoBERTa, BERTweet**	**Label flipping, Trigger injection, Clean-label poisoning**	**Sentiment140**	**Yes**

III. METHODOLOGY

A. Dataset

The Sentiment140 corpus [21] contains 1.6 million labelled English tweets. Labels are remapped to binary (0 = negative, 1 = positive) [7], [12]. A stratified random sample of N ∈ [200k, 300k] tweets are taken, preserving class balance. Preprocessing follows social-media NLP conventions [3]: @mentions and URLs are removed, text is lowercased, and emoticons and punctuation are retained. The sample is split 70/10/20 into train, validation, and test sets. All poisoning is applied exclusively to the training sets; the validation and test sets remain strictly clean.

B. Models

BERT-base-uncased [1] serves as the general-domain baseline. RoBERTa-base [2] provides a large-scale optimized alternative. BERTweet [3], pre-trained on 850 million tweets, offers the closest distributional fit to Sentiment140. The three models span general-domain, large-scale, and domain-specific pre-training, enabling analysis of how each dimension affects robustness.

C. Training Configuration

All models are implemented with Hugging Face Transformers [12] and PyTorch on a GPU-accelerated environment, fine-tuned under identical conditions (Table II). Each experimental cell (model × attack × α) is replicated under three random seeds; results are reported as mean ± SD [10], [15]. The complete end-to-end pipeline—from raw data through poisoning, fine-tuning, and evaluation—is illustrated in Fig. 1.

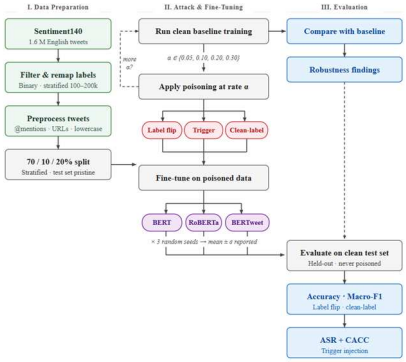


Fig. 1. Overall workflow of the experimental pipeline. The Sentiment140 corpus is pre-processed, stratified-sampled, and split 70/10/20; poisoning is applied only to the training set under three attack strategies (label flipping, trigger injection, clean-label) at α ∈ {0.05, 0.10, 0.20, 0.30}. Each (model × attack × α) condition is repeated under three random seeds and evaluated on the held-out clean test set.

TABLE II. FINE-TUNING HYPERPARAMETERS

Hyperparameter	Value	Justification
Learning rate	2e-5	Standard for BERT fine-tuning
Batch size	32	Gradient stability vs. memory
Epochs	3	Convergence without overfitting
Max sequence length	128	Covers >95% of tweet lengths
Optimizer	AdamW	Decoupled weight decay [8]
Warmup steps	10% of total	Stabilizes pre-trained weights

D. Poisoning Strategies

Three attack categories span naive label manipulation to covert feature-space perturbation. Poisoning rate α ∈ {0.05, 0.10, 0.20, 0.30} is applied to the training set; α = 0 is the clean baseline.

1) Label Flipping

A randomly selected subset S of training samples has ground-truth labels inverted via f(0) = 1, f(1) = 0

$$D' = (D \setminus S) \cup \{(x_i, f(y_i)) : i \in S\} \quad (1)$$

Because text is left unchanged, this attack is the most transparent and acts as the lower-bound disruptive baseline [4], [14].

2) Trigger Injection

A fixed rare token t (e.g., "cf") is inserted into each selected sample alongside a label flip:

$$x'_i = x_i \oplus t,\ y'_i = f(y_i)\ for\ i \in S \quad (2)$$

This follows BadNets-style backdoor attacks [27] applied to text [5], [13]. A successful backdoor is signalled by high ASR on triggered inputs with maintained CACC on clean ones.

3) Clean-Label Poisoning

Synonym substitution replaces strongly valanced tokens with weaker alternatives (e.g., "great" → "acceptable") to reduce the sentiment signal without label changes [16]. Character-level noise substitutes characters with visually similar variants (e.g., "loved" → "l0ved"), producing anomalous subword decompositions under WordPiece/BPE tokenization [17], [26], [30]. Both perturbations bypass label-consistency audits [6], [19].

E. Evaluation Metrics

Each adversarial strategy impacts model behaviour through a distinct mechanism, therefore, we deploy a multi-metric evaluation framework to capture these varied degradation profiles (Table III). Global and per-class performance drops are tracked via Accuracy and Macro-F1. Macro-F1 is chosen over micro-averaged counterpart to ensure each sentiment class receives equal weight that provides critical distinction when targeted poisoning forces a model's predictions to skew heavily toward a single target label [9]. To evaluate targeted manipulation, the Attack Success Rate (ASR) quantifies the exact proportion of trigger-bearing inputs that are successfully manipulated into the adversary's target class [13]. CACC is the model's accuracy on unperturbed inputs; high ASR combined with maintained CACC indicates a successful backdoor attempt [5], [13], [27].

TABLE III. EVALUATION METRICS PER ATTACK TYPE

Attack Type	Primary Metrics	Rationale
Label Flipping	Accuracy, Macro-F1	Global label corruption degrades overall performance [4]
Trigger Injection	ASR, CACC	Backdoor success requires high ASR with maintained CACC [5],[13]
Clean-Label Poisoning	Macro-F1, CACC	Covert feature perturbation degrades class boundaries without label anomalies [6]

IV. RESULTS AND DISCUSSION

To evaluate the impact of the attacks, we tested poisoning rates (α) at 0.00, 0.05, 0.10, 0.20, and 0.30. Every experiment was repeated using three random seeds to establish the average results and their standard deviations. Throughout our analysis, the unpoisoned run (α = 0.00) provides the clean baseline reference.

A. Clean Baseline

As shown in Table IV, the baseline results align with our initial expectations. BERTweet achieved the strongest performance, recording approximately 87.99% for both accuracy and Macro-F1. In comparison, RoBERTa and BERT yielded lower scores of 85.98% and 85.56%. The variance across the three random seeds was very minimal (SD ≤ 0.20 pp), which points to high experimental stability rather than just statistical chance. We also observed that this specific ranking among the models generally persisted even when adversarial attacks were introduced later in the study.

TABLE IV. CLEAN BASELINE PERFORMANCE (A = 0.00), MEAN ± SD ACROSS THREE SEEDS

Model	Accuracy (%)	Macro-F1 (%)
BERT	85.56 ± 0.14	85.56 ± 0.14
RoBERTa	85.98 ± 0.14	85.98 ± 0.15
BERTweet	87.99 ± 0.15	87.99± 0.15

B. Label Flipping Results

Looking at the data in Table V and Figure 2, performance steadily drops for all three models as the poisoning rate (α) goes up. However, introducing a 5% poisoning rate (α = 0.05) barely moves the needle. The scores shift by less than 0.2 percentage points from the baseline, and the standard deviation ranges still largely overlap with the unpoisoned results. It seems the transformer fine-tuning process is robust enough to naturally filter out this minor level of label corruption without taking a hit to overall accuracy.

True model divergence only manifests at α = 0.10. By α = 0.30, BERT has dropped 3.95 pp (84.52% to 80.76%), RoBERTa falls 2.77 pp (85.33% to 82.95%), and BERTweet

declines by 2.65 pp (87.63% to 85.31%). Crucially, BERT's performance dispersion expands significantly at this peak rate (SD = 1.24 pp), whereas RoBERTa and BERTweet whereas RoBERTa and BERTweet sustain highly stable distributions (SD ≤ 0.25 pp).

These results isolate three key trends. First, label flipping stands out as the unique adversarial vector capable of forcing a predictable, intensity-dependent degradation across all architectures. Second, BERTweet's mitigated degradation and tight seed variance validate the hypothesis that domain-specific pre-training infuses models with robust inductive priors that effectively buffer against label contamination. This results at the performance gulf separating baseline BERT from the more robust, domain-adapted models widens progressively as the dataset is increasingly poisoned.

TABLE V. LABEL FLIPPING – TEST ACCURACY (%), MEAN ± SD ACROSS THREE SEEDS

Model	α = 0.00	α = 0.05	α = 0.10	α = 0.20	α = 0.30
BERT	84.74 ± 0.11	84.62 ± 0.20	84.52 ± 0.14	82.62 ± 0.38	80.76 ± 1.24
RoBERTa	85.73 ± 0.45	85.54 ± 0.14	85.33 ± 0.12	84.60 ± 0.08	82.95 ± 0.25
BERTweet	87.94 ± 0.20	87.99 ± 0.10	87.63 ± 0.12	86.83 ± 0.18	85.31 ± 0.08

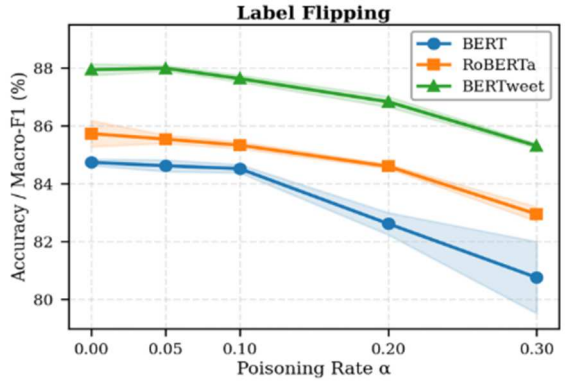


Fig. 2. Accuracy/Macro-F1 under label flipping. Shaded bands = ±1 SD.

C. Trigger Injection Results

Table VI reveals that ASR across all three models consistently hovers around 45 to 50% regardless of the poisoning rate. For a binary classification task, this range is essentially equivalent to random guessing. No clear trends emerge from the data, and in some cases intermediate poisoning rates produce ASR values below the clean baseline, as seen with BERTweet at α = 0.05 which recorded 45.53 ± 1.76%. Since the SD bands at every rate fully overlap with those of the clean baseline, these small fluctuations are better explained by statistical noise than by any meaningful backdoor association taking hold.

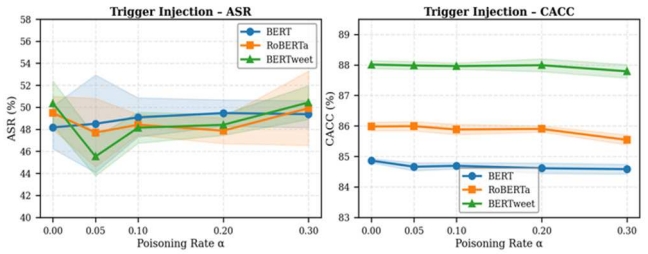
CACC also stays largely flat throughout, with the maximum drift reaching only 0.44 pp in RoBERTa at α = 0.30. The backdoor pattern commonly reported in prior literature, where high ASR coexists with maintained CACC [5], [27], was not replicated here.

Several factors likely explain this divergence. For one, subword tokenizers such as BPE and WordPiece tend to fragment or alter the trigger token in ways that reduce its expected rarity and distinctiveness. On top of that, restricting the trigger to a single insertion index probably provides a too narrow a contextual signal to meaningfully shift the final [CLS] representation. Lastly, a fine-tuning window of only three epochs appears to be too short for the network to internalize any hidden backdoor association in a reliable way.

TABLE VI. TRIGGER INJECTION – ATTACK SUCCESS RATE (ASR), CLEAN ACCURACY (CACC) (%), MEAN ± SD ACROSS THREE SEEDS

Model	Metric	α = 0.00	α = 0.05	α = 0.10	α = 0.20	α = 0.30
BERT	ASR	48.17 ± 1.94	48.50 ± 4.43	49.08 ± 1.77	49.47 ± 1.20	49.37 ± 1.23
BERT	CACC	84.86 ± 0.07	84.66 ± 0.13	84.69 ± 0.10	84.61 ± 0.17	84.58 ± 0.16
RoBERTa	ASR	49.48 ± 1.51	47.70 ± 3.10	48.41 ± 0.94	47.87 ± 1.18	49.91 ± 3.37
RoBERTa	CACC	85.98 ± 0.14	85.99 ± 0.15	85.88 ± 0.17	85.90 ± 0.10	85.54 ± 0.16
BERTweet	ASR	50.36 ± 2.03	45.53 ± 1.76	48.14 ± 1.41	48.40 ± 0.95	50.43 ± 1.53
BERTweet	CACC	88.01 ± 0.13	87.98 ± 0.13	87.96 ± 0.10	87.99 ± 0.21	87.79 ± 0.22

Fig. 3. ASR (left) and CACC (right) under trigger injection. ASR remains at chance level across all a and models.



D. Clean-Label Poisoning Results

Table VII shows negligible degradation under clean-label poisoning across all three models. RoBERTa showed the most uniform downward trend (Macro-F1: 85.97% → 85.34%, −0.63 pp), however the SD at higher α (0.45–0.53 pp) scales closely with mean degradation, the observed trend is only marginally distinct from noise. BERTweet declines only 0.19 pp; BERT shows no net degradation across all α (range: 84.58%–84.81%, within seed SD). These Macro-F1 trends are consistently aligned in the CACC results. The clean-label perturbations failed to shift decision boundaries: synonym substitutions that do not cross the polarity threshold leave sufficient contextual signal for fine-tuning to recover [23], and single-character substitutions are typically absorbed into familiar subword sequences by the tokenizer [17], [26].

The clean-label perturbations failed to shift decision boundaries: synonym substitutions that do not cross the polarity threshold leave sufficient contextual signal for fine-tuning to recover, and single-character substitutions are typically absorbed into familiar subword sequences by the tokenizer. However, it must be noted that this failure is primarily due to our use of non-adaptive synonym swaps. Modern clean-label attacks that leverage gradient-based feature collisions or latent-space perturbations are known to be far more effective. Therefore, this outcome indicates an attack configuration limitation rather than an inherent architectural immunity within the Transformers.

TABLE VII. CLEAN-LABEL POISONING – MACRO-F1 AND CACC (%) AND MEAN ± SD ACROSS THREE SEEDS

Model	Metric	α = 0.00	α = 0.05	α = 0.10	α = 0.20	α = 0.30
BERT	Macro-F1	84.74 ± 0.13	84.71 ± 0.19	84.71 ± 0.06	84.58 ± 0.15	84.81 ± 0.31
BERT	CACC	84.75 ± 0.11	84.72 ± 0.18	84.72 ± 0.06	84.59 ± 0.14	84.81 ± 0.31
RoBERTa	Macro-F1	85.97 ± 0.15	85.75 ± 0.45	85.55 ± 0.37	85.47 ± 0.53	85.34 ± 0.53
RoBERTa	CACC	85.98 ± 0.14	85.76 ± 0.45	85.55 ± 0.37	85.48 ± 0.53	85.35 ± 0.52
BERTweet	Macro-F1	87.99 ± 0.15	88.03 ± 0.16	87.88 ± 0.05	87.83 ± 0.17	87.80 ± 0.16
BERTweet	CACC	87.99 ± 0.15	88.03 ± 0.16	87.88 ± 0.05	87.83 ± 0.17	87.80 ± 0.17

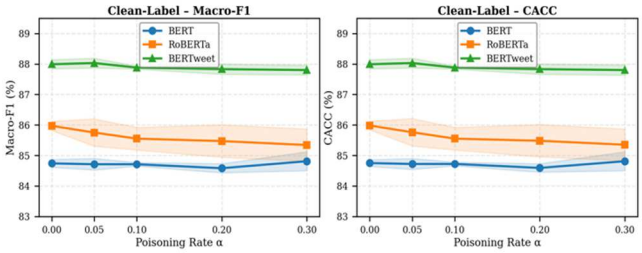


Fig. 4. Macro-F1 (left) and CACC (right) under clean-label poisoning. Drops remain below 1 pp for all models.

E. Cross-Model Comparison and Research Question Synthesis

(RQ1) Only label flipping produced a clearly monotonic, dose-dependent degradation curve, with absolute drops of 2.65–3.95 pp at α = 0.30 that exceed seed-level SDs by a wide margin. Trigger injection yielded no measurable backdoor signature (ASR ≈ 45–50%; CACC drift ≤ 0.44 pp), and clean-label drops were sub-percentage-point and often indistinguishable from seed variance. Label flipping is the sole statistically reliable threat in this experimental setting.

(RQ2) BERTweet consistently stayed ahead in accuracy at every α level, suffering the smallest overall drop (-2.65 pp). On the other end of the spectrum, the baseline BERT model took the heaviest hit (-3.95 pp) and showed the most instability across runs (SD = 1.24 pp at α = 0.30). It is particularly interesting that BERTweet ended up being more robust than RoBERTa, even with a smaller pre-training dataset. This points to the idea that training on domain-specific data acts as a natural buffer against noisy labels. Because the model already deeply understands social media text, a small batch of corrupted labels simply isn't enough to override its baseline knowledge.

(RQ3) Label flipping stands out as the most severe practical threat we tested within this operating range. The attack doesn't need to be clever or mathematically sophisticated; without defensive filters, bluntly corrupting the target labels causes the most damage. That said, the fact that trigger injection and clean-label poisoning didn't work here doesn't mean those methods are fundamentally weak. It is more likely that our specific attack configurations were just too mild to overcome the resilience of modern PLM fine-tuning. Exploring stronger, more advanced variations of these attacks [27], [29] should definitely be a priority for future studies.

F. Experimental Limitations

While our findings highlight important performance trends, several methodological limitations must be acknowledged. First, the reliance on a single dataset (Sentiment140) limits the generalizability of our findings to other sentiment analysis datasets and broader NLP tasks. Second, our evaluation is based on relatively weak and simplistic attack implementations, lacking stronger state-of-the-art methods like advanced textual backdoors, feature-collision attacks, or optimization-based clean-label attacks. Third, our analysis relies on mean ± standard deviation across three random seeds. This limited number of repetitions and the absence of formal statistical significance testing constrain our ability to draw definitive conclusions regarding true model stability. Finally, no defense mechanisms were experimentally evaluated, which reduces the practical security contribution of this study.

V. CONCLUSION

This study compared BERT, RoBERTa, and BERTweet under label flipping, trigger injection, and clean-label poisoning on the Sentiment140 corpus across five poisoning rates (α ∈ {0.00, 0.05, 0.10, 0.20, 0.30}) and three seeds per condition, with evaluation on a held-out clean test set throughout. Three principal findings emerge. The clean performance ranking BERTweet > RoBERTa > BERT (87.99 ± 0.15% > 85.98 ± 0.14% > 84.74 ± 0.14%) was reproduced, confirming experimental validity. Then, label flipping was the only attack that produced reliable, monotonic degradation: BERT declined the most (−3.95 pp at α = 0.30) and BERTweet the least (−2.65 pp), with BERTweet also exhibiting the narrowest seed variance—findings that directly support the hypothesis that domain-aligned pre-training provides robustness against label noise beyond what optimized general-purpose training achieves. Ultimately, our trigger injection and clean-label poisoning failed to degrade the models as expected. This shouldn't be misconstrued as some inherent architectural immunity built into Transformers; it simply means our attack parameters were too conservative to break them.

These poor results are explained by several strict constraints on the program. The trigger injection, first, was a static single-character bigram fixed at the same position in a short three-epoch fine-tuning window. This setup was probably too weak to establish a lasting backdoor relationship, especially since BPE and WordPiece tokenizers tend to fragment isolated characters. Similarly, our clean-label tweaks were too small for highly expressive transformer encoders. Swapping of isolated characters or dropping in of basic synonyms left a strong contextual signal that the fine-tuning process could easily bypass. Furthermore, while our maximum poisoning rate (α=0.30) is consistent with realistic threat models, the experiments ultimately revealed a weak adversarial setting rather than genuine model resilience.

Whether the robustness hierarchy observed in label flipping holds up under intense stress remains an open question. Testing this requires moving away from conservative baselines toward much harsher adversarial setups. Future work must explore more aggressive backdoor methods, including learned syntactic triggers [29], multi-token strings, and direct weight poisoning [5], combined with

optimized clean-label techniques like gradient-guided synonym selection and feature collision frameworks. We also need to scale our testing suite to larger encoders like RoBERTa and BERTweet, alongside instruction-tuned variants. Evaluating these models against broader benchmarks like TweetEval and SemEval—while introducing defenses like activation clustering [31], spectral signatures [32], and certified bounds [4]—will give us a much more realistic view of current security limits.

This work emphasizes an important methodological caution. Statements about adversarial vulnerability depend heavily on the attack intensity chosen. To avoid making incorrect conclusions about differences in structural models, future comparisons need to clearly adjust and test attack parameters. Only then can we fairly attribute performance gaps to fundamental architectural properties.

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AUTHOR CONTRIBUTORSHIP

Clarencia Augustine took charge of writing the full manuscript, handling everything from the opening chapter to the conclusion, while also assembling the final document. The study's core research topic was conceptualized by Devon Nicholas, who also ran all experiments, built the system diagrams, and wrote the Methodology and Reference sections. Vania Alicia Yocelin organized the Related Works review and helped manage the citations. Finally, Fiqri Ramadhan Tambunan alongside Meiliana guided the research as supervisors, reviewing the text and making essential edits to refine the manuscript.

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